Can Oluk

Department of Psychological & Brain Sciences
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Interests: visual perception, computational neuroscience, ideal-observer and image-computable models.

Appointments

Postdoctoral Researcher, Boston University — 2026 – Present
Principal Investigator: Prof. Rachel Denison

Postdoctoral Researcher, Brain and Mind Institute, EPFL — 2024 – 2026
Time course of (conscious) visual processing
Principal Investigator: Prof. Michael Herzog

Postdoctoral Researcher, Laboratoire des systèmes perceptifs, ENS — 2023 – 2024
Confidence in motion perception
Principal Investigator: Prof. Pascal Mamassian

Education

Ph.D. Psychology, The University of Texas at Austin — 2016–2022
Completed the Center for Perceptual Systems interdisciplinary training program
Supervisor: Prof. Wilson S. Geisler

B.A. Psychology, Minor in Philosophy, Bilkent University — 2011–2016

Research Experience

Graduate Research Assistant, Center for Perceptual Systems, U.T. Austin — 2016 – 2022
Slant perception, Target detection under uncertainty
Supervisor: Prof. Wilson S. Geisler

Undergraduate Thesis Student, UMRAM, Bilkent University — 2015 – 2016
Multiplexed echo planar imaging (fMRI) methods
Supervisor: Prof. Huseyin Boyaci

Undergraduate Research Assistant, UMRAM, Bilkent University — 2013 – 2016
Audiovisual associations, Rapid Motion aftereffects
Supervisor: Prof. Hulusi Kafaligonul

Publications

Journal Articles

Oluk, C., and Geisler, W. S. (2025). Target identification under high levels of amplitude, size, orientation and background uncertainty. *Journal of Vision*, 25(2):3. [PDF] [Code/Data]

Oluk, C., and Geisler, W. S. (2023). Effects of Target-Amplitude and Background-Contrast Uncertainty Predicted by a Normalized Template-Matching Observer. *Journal of Vision*, 23(12):8. [PDF]

Oluk, C., Bonnen, K., Burge, J., Cormack, L. K., and Geisler, W. S. (2022). Stereo slant discrimination of planar 3D surfaces: Frontoparallel versus planar matching. *Journal of Vision*, 22(5), 6-6. [PDF] [Code/Data]

Oluk, C., Pavan, A., and Kafaligonul, H. (2016). Rapid motion adaptation reveals the temporal dynamics of spatiotemporal correlation between ON and OFF pathways. *Scientific Reports*, 6(1), 1-10. [PDF]

Kafaligonul, H. and **Oluk, C.** (2015). Audiovisual associations alter the perception of low-level visual motion. *Frontiers in Integrative Neuroscience*, 9, 26.

Manuscripts in Preparation

Oluk, C., Szinte, M., Masson, G. S., and Mamassian, P. (in prep). Confidence responses in global motion discrimination task are well predicted by visual reliability.

Book Chapters

Oluk, C., and Herzog, M. H. (2026). Discreteness versus continuity of conscious perception. In C. van Leeuwen (Ed.), *Comprehensive Consciousness* (2nd ed.). Elsevier.

Conference Abstracts

Holdsworth, M., Menétrey, M. Q., **Oluk, C.**, and Herzog, M. H. (2026). Evidence for winner-take-all dynamics in long-lasting visual integration. *Vision Science Society Meeting*, Florida, US.

Oluk, C. and Geisler, W. S. (2025). Evaluating ideal observers for large target identification tasks under additive white noise. *77th European Conference on Visual Perception*, Mainz, Germany.

Holdsworth, M., Menétrey, M. Q., **Oluk, C.**, and Herzog, M. H. (2025). A continuous adjustment task allows for better understanding of visual feature integration in a Sequential Metacontrast Paradigm. *77th European Conference on Visual Perception*, Mainz, Germany.

Oluk, C. and Geisler, W. S. (2025). Maximum likelihood estimates in visual categorization tasks are biased when pairwise discriminabilities are imbalanced. *Systems Vision Science Symposium*, Tübingen, Germany.

Oluk, C., Szinte, M., Masson, G. S., Mamassian, P. (2024). Confidence responses in global motion discrimination task are well predicted by visual reliability. *Vision Science Society Meeting*, Florida, US.

Oluk, C. and Geisler, W. S. (2021). The Energy-Normalized MAX Observer Approximates the Ideal Observer Under High-levels of Simultaneous Orientation and Scale Uncertainty in White Noise. *Vision Science Society Meeting*, Florida, US.

Oluk, C. and Geisler, W. S. (2020). Ideal Observers for the estimation of disparity in random-pixel stereograms. *Vision Science Society Meeting*, Florida, US.

Oluk, C., and Geisler, W. S. (2019). Effects of Target Amplitude Uncertainty, Background Contrast Uncertainty, and Prior Probability Are Predicted by the Normalized Template-Matching Observer. *Vision Science Society Meeting*, Florida, US.

Oluk, C., Bonnen, K., Burge, J., Cormack, L., and Geisler, W. (2018). Stereo Slant Estimation of Planar Surfaces: Standard Cross-Correlation vs. Planar-Correlation. *Vision Science Society Meeting*, Florida, US.

Kafaligonul, H. and **Oluk, C.** (2014). Audiovisual associations alter the perception of low-level visual motion. *Annual Meeting of the Society for Neuroscience*, Washington, D.C., US.

Kafaligonul, H. and **Oluk, C.** (2014). Altering perception of low-level visual motion by audiovisual associations. *37th European Conference on Visual Perception*, Belgrade, Serbia.

Invited Talks

"Finding Scalable Principles of Vision: Computational Mechanisms of 3D Surface Slant Estimation"
Department of Neuroscience, Bilkent University April 2026

"Towards Scalable Computational Principles of Perception: Modeling Human Behavior Under Realistic Task Complexity"
Department of Psychology, Bilkent University February 2026

"Confidence in Global Motion Direction Discrimination"
Centre IRM - INT, Institut de Neurosciences de la Timone, Aix-Marseille Université 2024

Grants

Principal Investigator, TUBITAK 2209/A 2015 – 2016
"Associative Learning and Motion Induced Plasticity"
Supervisor: Prof. Hulusi Kafaligonul
Competitive national undergraduate research grant covering project costs.

Honours and Awards

Lloyd A. Jeffress Memorial Fellowship 2016, 2019

Bilkent University Comprehensive Scholarship 2011 – 2016
Full tuition waiver and stipend

Teaching Experience

Guest Lecturer, EPFL Spring 2025
BIO-483 - Neuroscience: behavior and cognition
Three-hour lecture on attention and consciousness

Teaching Assistant, U.T. Austin Spring 2019
PSY 323 - Perception

Teaching Assistant, Bilkent University Fall 2013
CS 121 - Introduction to Computer Tools

Supervision & Mentoring

MSc thesis co-supervision, Université de Strasbourg (hosted at EPFL) 2024 – 2025
Determinants of the duration of window of integration during the temporal processing of visual stimuli

Intern supervision, EPFL Spring 2025
Attention in Sequential Metacontrast Paradigm

Undergraduate project mentor, EPFL Spring 2025
TAFKAP-Probabilistic Analysis of Cortical Activity in fMRI.

Service & Memberships

Ad Hoc Reviewer — Journal of Vision (recognized "Exceptionally Good Review," 2023)

Ad Hoc Reviewer – Attention, Perception, & Psychophysics

Vision Science Society 2017–Present

Miscellaneous

Languages: English (advanced), Turkish (native)

Programming Skills: Matlab, Python, LaTeX, SPSS, Stata

Last updated: August 5, 2026